

The chapter in one picture: three effects, three scales, three toolboxes

multipath fading

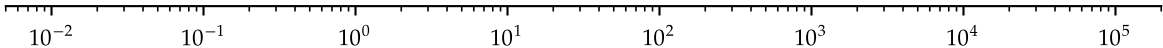
20--40 dB swings every $\lambda/2$
equalizers, OFDM, diversity, codes

shadowing

4--12 dB, log-normal
margins, overlap, handover

path loss

d^{-2} to d^{-4}
link budget, antennas, cell size



distance over which the effect changes (m), at about 2 GHz